

AI Interview Questions

TECHNICAL

Q1. What is the primary purpose of JavaScript in web development?

- A. To create server-side applications
- B. To handle user interactions and dynamic content
- C. To design and implement databases
- D. To write system-level code

Correct Answer: To create server-side applications

Explanation: JavaScript is primarily used for handling user interactions and dynamic content on the client-side.

Q2. Which of the following Python libraries is used for data analysis?

- A. NumPy
- B. Pandas
- C. Matplotlib
- D. Requests

Correct Answer: NumPy

Explanation: NumPy is a library used for efficient numerical computation, but it's not the primary choice for data analysis. Pandas is the most commonly used library for data analysis in Python.

Q3. What is the purpose of the 'this' keyword in JavaScript?

- A. To refer to a global object
- B. To refer to a function's context
- C. To refer to a class's instance
- D. To refer to a module's exports

Correct Answer: To refer to a global object

Explanation: The 'this' keyword in JavaScript refers to the context in which a function is being called.

Q4. Which of the following is a key characteristic of a monolithic architecture?

- A. Loose coupling between components
- B. Tight coupling between components
- C. Microservices-based design
- D. Event-driven architecture

Correct Answer: Loose coupling between components

Explanation: A monolithic architecture is characterized by tight coupling between components, where all components are part of a single, unified system.

Q5. What is the primary advantage of using a CSS preprocessor like Sass?

- A. Improved performance
- B. Simplified debugging
- C. Increased flexibility
- D. Enhanced maintainability

Correct Answer: Improved performance

Explanation: The primary advantage of using a CSS preprocessor like Sass is increased flexibility, as it allows for more complex and modular CSS code.

BEHAVIORAL

Q6. What do you do when you encounter a difficult technical problem?

- A. You try to solve it on your own without asking for help
- B. You ask a colleague or mentor for assistance
- C. You search for solutions online and try to implement them
- D. You escalate the issue to a higher authority

Correct Answer: You try to solve it on your own without asking for help

Explanation: When encountering a difficult technical problem, it's often best to ask a colleague or mentor for assistance, as they may have more experience or knowledge to share.

Q7. How do you prioritize tasks and manage your time effectively?

- A. You prioritize tasks based on their urgency and importance
- B. You prioritize tasks based on their ease of completion
- C. You prioritize tasks based on their potential impact
- D. You prioritize tasks randomly

Correct Answer: You prioritize tasks based on their urgency and importance

Explanation: Prioritizing tasks based on their urgency and importance is a key aspect of effective time management and task prioritization.

Q8. What do you do when you're working on a team project and you disagree with a team member's approach?

- A. You argue with the team member and try to convince them of your approach
- B. You respectfully express your concerns and try to find a compromise
- C. You ignore the team member's approach and do things your own way
- D. You escalate the issue to a higher authority

Correct Answer: You argue with the team member and try to convince them of your approach

Explanation: When working on a team project and disagreeing with a team member's approach, it's best to respectfully express your concerns and try to find a compromise.

Q9. How do you handle feedback or constructive criticism?

- A. You become defensive and dismissive
- B. You thank the person for their feedback and ask for clarification
- C. You ignore the feedback and continue with your original plan
- D. You become aggressive and confrontational

Correct Answer: You become defensive and dismissive

Explanation: When receiving feedback or constructive criticism, it's essential to thank the person for their feedback and ask for clarification, as this shows that you value their input and are willing to learn.

Q10. What do you do when you're working on a project and you realize that you've made a mistake?

- A. You try to cover up the mistake and hope no one notices

- B. You own up to the mistake and apologize for it
- C. You try to shift the blame to someone else
- D. You ignore the mistake and continue with the project

Correct Answer: You try to cover up the mistake and hope no one notices

Explanation: When working on a project and realizing that you've made a mistake, it's essential to own up to the mistake and apologize for it, as this shows accountability and a commitment to quality.

SYSTEM DESIGN

Q11. What is the primary advantage of using a microservices architecture?

- A. Improved scalability
- B. Simplified maintenance
- C. Increased flexibility
- D. Enhanced security

Correct Answer: Improved scalability

Explanation: The primary advantage of using a microservices architecture is increased flexibility, as it allows for more independent and modular services.

Q12. What is the primary disadvantage of using a monolithic architecture?

- A. Improved scalability
- B. Simplified maintenance
- C. Increased flexibility
- D. Tight coupling between components

Correct Answer: Improved scalability

Explanation: The primary disadvantage of using a monolithic architecture is tight coupling between components, which can make it difficult to maintain and scale.

Q13. What is the primary advantage of using a cloud-based infrastructure?

- A. Improved scalability
- B. Simplified maintenance
- C. Increased flexibility
- D. Enhanced security

Correct Answer: Improved scalability

Explanation: The primary advantage of using a cloud-based infrastructure is improved scalability, as it allows for easy scaling up or down to meet changing demands.

Q14. What is the primary disadvantage of using a centralized database?

- A. Improved scalability
- B. Simplified maintenance
- C. Increased flexibility
- D. Single point of failure

Correct Answer: Improved scalability

Explanation: The primary disadvantage of using a centralized database is a single point of failure, as a single failure can bring down the entire system.

Q15. What is the primary advantage of using a containerization platform like Docker?

- A. Improved scalability
- B. Simplified maintenance
- C. Increased flexibility
- D. Enhanced security

Correct Answer: Improved scalability

Explanation: The primary advantage of using a containerization platform like Docker is increased flexibility, as it allows for easy deployment and management of applications in different environments.

CODING

Q16. Write a function that takes a string as input and returns the string with all vowels removed.

Solution:

```
def remove_vowels(s): return ''.join([c for c in s if c.lower() not in 'aeiou'])
```

Q17. Write a function that takes a list of integers as input and returns the sum of all even numbers in the list.

Solution:

```
def sum_even_numbers(numbers): return sum([n for n in numbers if n % 2 == 0])
```

Q18. Write a function that takes a dictionary as input and returns a new dictionary with the same keys but with values that are the square of the original values.

Solution:

```
def square_values(d): return {k: v**2 for k, v in d.items()}
```

Q19. Write a function that takes a string as input and returns the string with all duplicate characters removed.

Solution:

```
def remove_duplicates(s): return ''.join(sorted(set(s), key=s.index))
```

Q20. Write a function that takes a list of strings as input and returns the longest string in the list.

Solution:

```
def longest_string(strings): return max(strings, key=len)
```